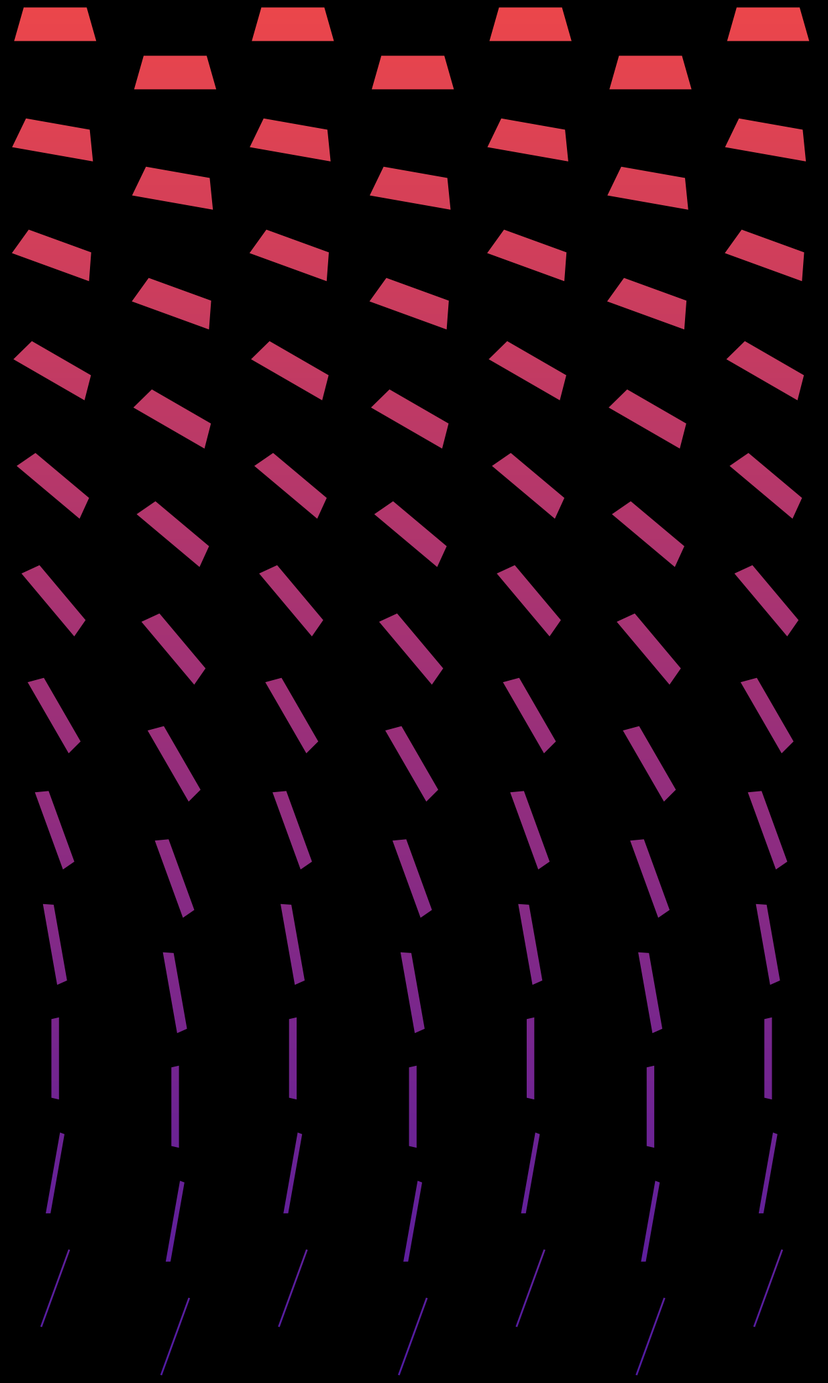
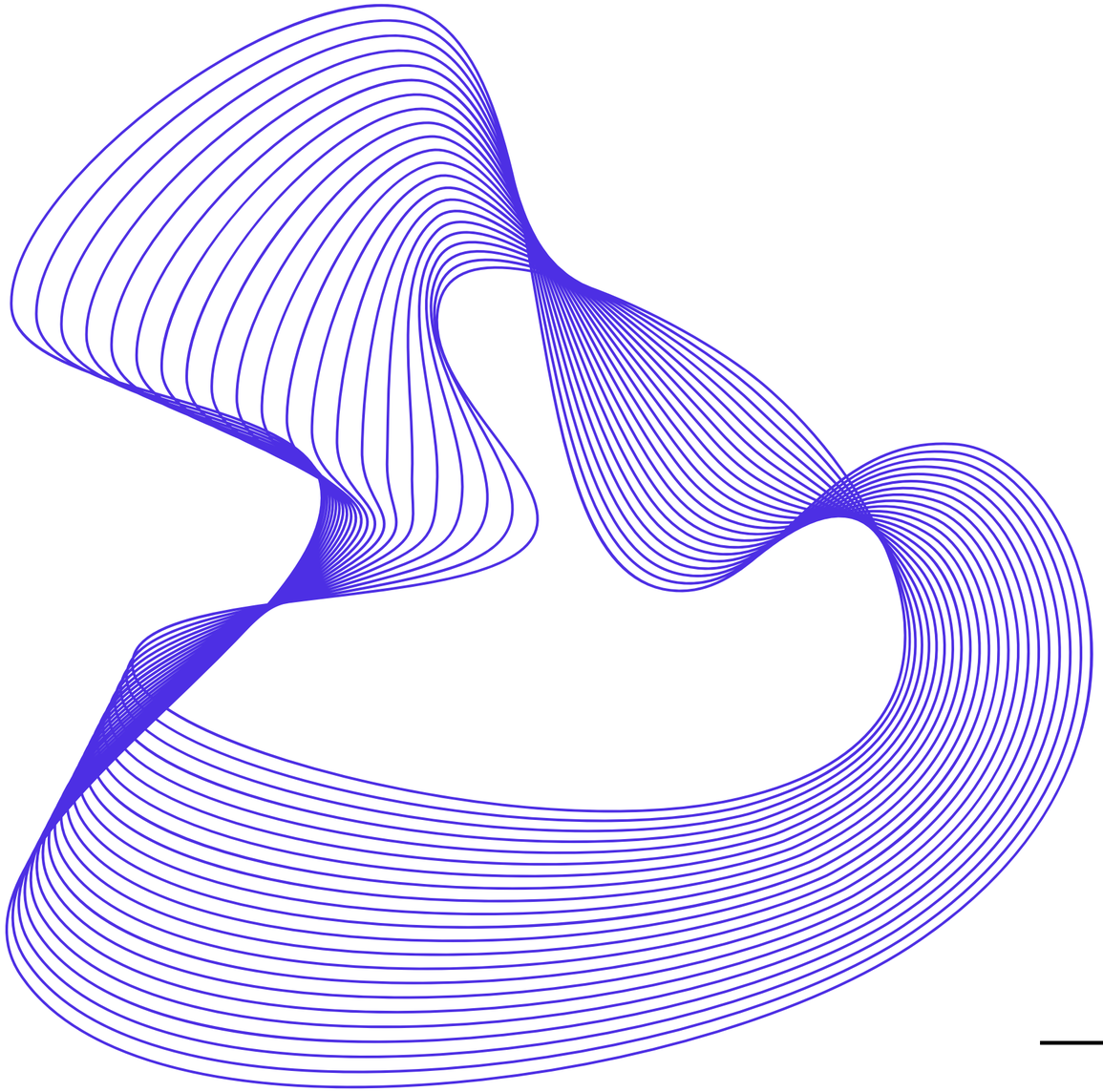
Unsupervised Learning in Python

Learning without labels





Clustering for dataset exploration

Unsupervised Learning

- Unsupervised learning finds patterns in data
- E.g., *clustering* customers by their purchase
- Compressing the data using purchase patterns (dimension reduction)

Unsupervised Learning

- Learn without labels
- Pure pattern discovery
- Without a specific prediction task in mind.

k-means clustering

- Finds clusters of samples
- Number of clusters must be specified
- Implemented in *scikit-learn*

Evaluating a cluster

- If creating a new clustering algorithm, use complete datasets to check correspondence with the labels

pandas

pandas.crosstab()

- Measure the quality of the clustering and use this information to choose the number of clusters (hyper parameter *k*)

scikit learn

model.inertia_

How does KMeans work?

Algorithm 1 k-means algorithm

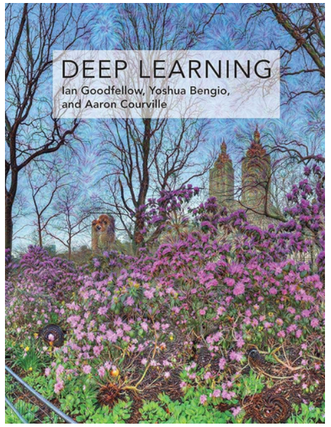
1: Specify the number k of clusters to assign.
2: Randomly initialize k centroids.
3: **repeat**
4: **expectation:** Assign each point to its closest centroid.
5: **maximization:** Compute the new centroid (mean) of each cluster.
6: **until** The centroid positions do not change.



K-Means Clustering in Python: A Practical Guide

In this step-by-step tutorial, you'll learn how to perform k-means clustering in Python. You'll review evaluation metrics for choosing an appropriate number of clusters and build an end-to-end k-means clustering pipeline in scikit-learn.

realpython



Source: https://www.deeplearningbook.org/contents/ml.html

Using KMeans

```
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

  define the model hyperparameters
model = KMeans(n_clusters=  )

  train the model (replace <samples> with your data)
model.fit(samples)

  predict the labels (cluster number) from the same samples. Can also predi
labels = model.predict(samples)
print(labels)

 scatter plot to see data points in color by cluster label
xs = samples[:,  ]    choose the first feature column
ys = samples[:,  ]    choose the third feature column
plt.scatter(xs, ys, c=labels, alpha=  .  , )

  Assign the cluster centers: centroids
centroids = model.cluster_centers_

  Assign the columns of centroids: centroids_x, centroids_y
centroids_x = centroids[:,  ]
centroids_y = centroids[:,  ]

  Make a scatter plot of centroids_x and centroids_y
plt.scatter(centroids_x, centroids_y, marker='D', s=    )

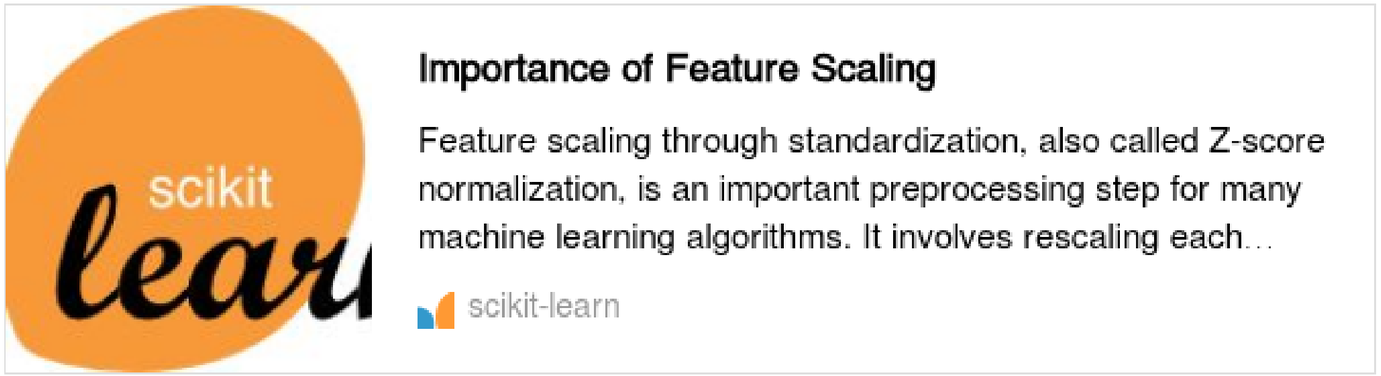
plt.show()
```

using_KMeans.py hosted with ❤ by GitHub view raw

Cluster labels for new samples

- New samples are assigned to existing clusters without starting over
- k-means remembers the mean (or average) of the samples in each cluster (a.k.a. ‘centroids’)
- New samples are assigned to the cluster whose centroid is closest

Evaluating a cluster



Lab: Mall customers dataset

1. Download the mall_customers.csv dataset.

2. Find the best value of k using the elbow method. This method involves running the KMeans clustering algorithm for a range of k values and then plotting the number of clusters against the sum of the squared distances within each cluster (inertia). The "elbow" in this plot, i.e. the inflection point where the rate of decline (angle) becomes significantly lower, may indicate a good value of k.

3. With the new customer groups, which one would you focus on?, and what actionable marketing strategy can you recommend?

4. Finally, in which group would you place a new customer with an annual income (k$) of 110, and spending score of 21?

Estimated time: 30min. After the 30min, student groups will present their findings.



Thank you

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